# Bijux Pollenomics: scientific evidence

## Sources, method correspondence, predictions and model use

Bijux Pollenomics

30 September 2026

Source-bound research briefing

# Evidence ladder

1. Archive/source custody and taxon-count mapping.
2. Author-equation and numerical implementation parity.
3. Observed-pollen held-out prediction.
4. LRA reconstruction and forward-pollen prediction.
5. Independent measured-vegetation validation: no admitted pair yet.

Each rung answers a different question; a result at one rung does not imply the next.

Evidence: `reports/README.md`; `reports/needed/README.md`

# Four-country source and prediction accounting

| Target | Sites | Samples | Assigned/native NISP | Scored/attempted bins |
|---|---|---|---|---|
| Denmark | 1 | 91 | 122,998/129,768 | 24/24 |
| Finland | 28 | 1,797 | 1,229,019/1,260,281 | 435/435 |
| Norway | 52 | 3,544 | 2,018,295/2,286,020 | 1,084/1,178 |
| Sweden | 3 | 228 | 221,800/240,966 | 51/54 |

Closed 31-target author pollen composition. Excluded/unmapped mass remains outside the score. Zero Nordic samples are admitted to LRA by this study.

Evidence: `reports/science/denmark-finland-norway-author-pollen.md; reports/science/sweden-author-pollen.md`

# Swedish training scope on shared pollen targets

- Hold out each Swedish site; train on the other two Swedish sites or those same two plus 81 DK/FI/NO sites.
- On 39 shared site-bins, inverse-distance-squared equal-site pollen MAE is 0.032846 versus 0.023226, favoring four-country training.
- Sweden-only refuses 15 of 54 candidate bins; four-country training predicts all 54.

Three previously exposed sites; most mean gain comes from Abborrtjarnen. No LRA or independent vegetation accuracy follows.

Evidence: `reports/science/sweden-training-scope.md`; `docs/report/regions/nordic/swedish-training-scope/paired-scores.csv`

# Swedish source support changes the pollen score

- We froze 1,419 method predictions across 473 supported queries; 13 of 486 site-bin/policy queries refused.
- Adjacent nominal 500-year source bins and a nearest-five cap changed IDW error slightly and differently by site.
- Excluding source centers within 100 km raised matched equal-site IDW MAE from 0.023151 to 0.029171 on 52 bins.

Three exposed Swedish sites; site centers, no age posterior, LRA reconstruction or measured vegetation.

Evidence: `reports/science/sweden-spatiotemporal-sensitivity.md`; `docs/report/regions/nordic/swedish-support-sensitivity/paired-scores.csv`

# What reproduces numerically

- Author raw-to-prepared replay: 17 example files, 871 raw sample rows, 15,028 matching prepared cells.
- WHH05: 301 raw grains map to 297 prepared; 31 paired means and SEs agree at 1500 BP.
- Reexecuted author R/native calculations on two grids (9+8 sites); 14,074 supported mean and SE values agree within declared tolerances.
- Archive TW2 differs: total variation 0.092/0.111; two taxa explain 73%/94% of each grid's L1 gap. Historical inputs remain unclosed.
- Archived ISTS and TBS mean arithmetic closes over all 10,500 grid-window rows, but biological classification and SE transformation remain open.

Evidence: `reports/science/publication-correspondence.md`; `reports/science/publication-correspondence/published-window-provenance.md`; `reports/science/publication-correspondence/author-example-reexecution.md`

# English LOVE: declared model profile

- 15 source sites, 1250–1750 BP, 30 targets; published PPE and fall speeds held fixed for retained taxa.
- Regional REVEALS plus ring-taxon-preserving LOVE; closed, nonnegative local cover.
- Uniform-start SLSQP: 300 iterations, 16,000 evaluations, objective tolerance $10^{-8}$.
- 192/195 training fits complete; three failures also refused under a bounded 600-iteration check.

These are source-specific development choices, not universally optimal LRA parameters or the original R optimizer.

Evidence: `reports/science/english-lra-pollen-prediction.md`

# English LOVE routes have different support

| Route | Source axis and attempts | Retained result |
|---|---|---|
| Analytical | 31 taxa; 15 folds | 15/15 invalid compositions; all 54 local ring rows include negative cover. |
| Differential evolution | 30 taxa; one pair, two requests | Both requests refused convergence; no held-out score. |
| Constrained SLSQP | 30 taxa; 15 folds | 14/15 folds predicted; BECKHAM had three training-fit refusals. |

Excluding 159 Filipendula counts changed the taxon response. These development runs are not a paired solver ranking or a vegetation-accuracy comparison.

Evidence: `reports/science/english-lra-pollen-prediction.md`

# Matched pollen comparison: result and limitation

| | MAE | | Hellinger | |
|---|---|---|---|---|
| | Direct | LRA | Direct | LRA |
| Training mean | .020496 | .026273 | .282082 | .360671 |
| Nearest | .019552 | .021132 | .255767 | .286001 |
| Inverse distance squared | .018579 | .021088 | .243673 | .285415 |

14 paired English sites, 30 taxa; all three direct means remain lower after any one site is omitted. Prior exposure and unequal training information limit interpretation.

Evidence: `docs/report/pollen-reconstruction/author-lra-constrained/direct-pollen/paired-hellinger.json`

# English basin radius: forward sensitivity only

- We replayed 42 frozen LRA vegetation estimates at 80%, 100% and 120% of each assumed target-lake radius.
- The actual forward consumer reproduced all 42 original pollen predictions exactly before target scoring.
- Inverse-distance-squared site-macro pollen MAE: .021032, .021088 and .021457 across those radii.
- On matched targets, direct pollen retains lower mean MAE and Hellinger for every method and radius factor.

14 prior-exposed English sites. Vegetation, regional cover and PPE were fixed; this does not calibrate basin geometry or admit Nordic LRA inputs.

Evidence: `docs/report/pollen-reconstruction/author-lra-constrained/radius-sensitivity/paired/report.json`; `reports/science/english-lra-pollen-prediction.md`

# Model selection follows the target and support

- Fixed spatial baselines permit clear paired interpretation on observed pollen.
- A bounded RBF candidate improved one prior-exposed author comparison on common support but reduced coverage.
- Country-excluded tuning did not establish superiority over inverse-distance-squared in the Nordic development holdouts.
- A two-case synthetic PPE/transport winner reflects a transport-law mismatch and is not a real-core parameter recommendation.
- For LRA, input identity, chronology, basin geometry, PPE and denominator must be qualified before any parameter search is meaningful.

Evidence: `reports/science/model-use.md`; `reports/science/synthetic-model-selection/README.md`; `reports/needed/README.md`

# HYDE and vegetation: retain the boundary

- Four-country HYDE source cropland grid sums match 84 country/year/scenario totals within $8.51 \times 10^{-11}$ km$^2$.
- This checks modeled-source internal accounting, not pollen-model benefit.
- The retained release has conflicting rights and version declarations; formal CRS and historical calendar support remain unresolved.
- No independent vegetation measurement pair has been admitted for prediction accuracy.

Next experiments require source semantics and a matched ecological reference, with outcomes protected from model selection.

Evidence: `reports/science/hyde-cropland-source-account.md`; `reports/needed/README.md`

# Open source decisions

- Abborrtjärnen: LandClim-selected age model, historical basin radius, eligible denominator, PPE use and rights.
- Bjäresjösjön: modern lake identity and nearby SEAD display intervals close; pollen-core identity and ecological event join remain refused.
- Other Nordic cores: physical-site and sample/window lineage before LRA admission.
- Historical publication: exact source revisions and authenticated PFT/uncertainty transformation.
- Independent confirmation: new qualified cohort frozen before target inspection.

Evidence: `reports/needed/README.md`; `reports/science/bjaresjo-context.md`